

Research Article

A fast and robust quantitative time-lapse assay for cell migration

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Abstract

We describe a simple and widely applicable method to measure cell migration in time-lapse sequences of fluorescently labeled cells in culture. Briefly, binarized cell images obtained after thresholding were cumulatively projected, and the covered areas were measured. This procedure determines the time course of the track area successively covered by the cell population. Under conditions where cell growth is negligible, a robust index of cell motility is derived from normalized plots for the displacement of cells over time. We applied this method to quantitatively examine the migration of B35 neuroblastoma cells transiently expressing GFP and to C6 glioma cells after staining with Hoechst 33258. This sensitive assay detected the influence of agents which inhibit actin polymerization (cytochalasin B) or interfere with the maintenance of cell polarity (methyl-beta-cyclodextrin) on cell migration. Thus, this assay is a versatile tool to measure quickly the migration of different cell types using different labeling strategies.

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Introduction

The migration of eukaryotic cells is a fundamental cellular process and involves the organization of cell body, plasma membrane and the underlying cytoskeleton into front and rear edges. At the molecular level, cell migration is based on the coordinated remodeling of the cytoskeleton which involves the polymerization of F-actin at the leading edge as force-generating process. To transmit the generated force to the surroundings, cells use specific adhesion receptors towards substrates or to adjacent cells [1].

Cell migration is of crucial importance during development of multicellular organisms and is involved in the normal physiology of various organ systems in the adult.

Moreover, migration processes are implicated in wound healing and many disease states including tumor metastasis and inflammation. Therefore, it is not surprising that numerous approaches were developed in the past to observe and measure the movements of cells. Ideally, one would be interested to develop a system which allows to follow the migration of single cells in intact tissue. This ideal has been realized, for example, in time-lapse imaging of the migration of melanoma cells in the skin of living mice using two-photon microscopy [2] and in selected small organisms such as early developing embryos, and in particular, zebrafish [3–5]. However, it is obvious that such or similar approaches are difficult to transfer to the study of tissues which are not visible from the body surface, such as, e.g. the mammalian central nervous system. In the latter case, slice cultures of brain tissue offer one possibility to study the behavior of cells in their tissue context. Time-lapse studies using slice cultures have provided important insights, e.g. into the migration of neuron precursors during forebrain development [6]. A simpler way to examine cell motility is the

Abbreviations: b-img, binarized image; BPA, black pixel area; img, image; Prj, projection; RD, relative displacement; RDI, relative displacement index.

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observation of dissociated cells *in vitro*. This type of experiment has been pioneered in studies on leukocyte movements [7] and was also used successfully with other cells, e.g. of epithelial origin [8]. An advantage of this approach is the possibility to manipulate the environment of the cells. For quantitative analyses, individual cells can be tracked in time-lapse sequences. This is done either manually (Jeffrey Kuhn: Multitracker, <http://rsb.info.nih.gov/ij/plugins/multitracker.html>) or using an automatic cell tracking algorithm (Nico Stuurman: MTrack2, <http://valelab.ucsf.edu/~nico/IJplugins/MTrack2.html>). From the constructed tracks, the migration speed, the migration persistence and other parameters can be calculated for each cell [9]. The distributions of these parameters under different conditions are then compared with the aid of statistical methods.

Since live-cell imaging microscopes are available which are equipped with motorized x – y stages, it is now possible to record simultaneously up to one hundred image sequences in a single experiment. This poses the challenge to develop fast methods of data analysis and to reduce the complexity of information inherent in these image sequences without precluding more detailed subsequent analyses of the same raw data. In the past, population approaches were used to examine cell migration when large numbers of conditions were to be compared. The classical method is the transwell chamber assay [10], where cells are placed onto a porous membrane and the number of cells which passed this membrane is quantified after a certain time interval. However, the process of migration itself cannot be observed in this assay. Other assays have been developed to derive simple numbers reflecting the migration behavior of cell populations such as the agarose drop assay. In this assay, cells are embedded in a droplet of low-melting agarose, and the distance cells migrated from the margin of the agarose drop is measured [11]. Recently, a cell migration assay using “high content screening” was developed where the area which is cleared from a substrate-coated fluorescent marker is used as measure for cell migration (Cellomics Inc.; <http://www.cellomics.com>).

All assays mentioned possess their specific advantages and shortcomings. Here, we describe a simple and general data analysis strategy to calculate a quantitative measure of migration for populations of cells from time-lapse experiments. This method is useful to screen a large number of conditions in time-lapse sequences of dissociated cells. This approach may be adapted easily to study cell migration in a wide range of experimental systems.

Materials and methods

Cell culture

B35 neuroblastoma cells [12] and C6 glioma cells [13] were grown in DMEM/F12 (Invitrogen) supplemented with

5% fetal calf serum, 0.3 g/l L-glutamine, 100 U/ml penicillin and 100 µg/ml streptomycin. B35 neuroblastoma cells were transfected using Exgene 500 (Fermentas) according to the instructions of the manufacturer. To stain living C6 glioma cells with Hoechst 33258, a stock solution in water (2% wt/vol) was diluted in culture medium 1:1000 and added to the cells.

Time-lapse video microscopy

For time-lapse studies, cells were plated in 24-well plates (Falcon, Becton-Dickinson) at the day prior to transfection. After transfection, the cells were grown overnight. Cell culture plates were mounted on a Zeiss Axiovert 200 epifluorescence microscope (Zeiss, Jena, Germany) equipped with a digital CCD camera (Axiocam, Zeiss), a motorized x – y stage (Scan IM 120 × 100; Märzhäuser GmbH, Germany), an automatic shutter and an *in vivo* incubation chamber (Incubator XL-3, Pelon GmbH, Germany). Images were recorded at low magnification (A-Plan 10x/0,25 Ph1, Zeiss) with a sampling interval of 15 min (Δt). The exposure time was kept constant for all positions and all time points. Images were processed with Axiovision 4.1 (Zeiss), exported in TIFF format and further analyzed using NIH Image J Version 1.32 (<http://rsb.info.nih.gov/ij/>).

Data analysis

The basic idea of the method is illustrated in Fig. 1. Stacks of the individual TIFF-images were generated in Image J using the command “File > Import > Image Sequence”. The single images (abbreviated *img*) were transformed into a binary black and white representation (abbreviated *b-img*) using thresholding with the same minimum gray-scale threshold value and for all images acquired in the experiment (command: “Image > Adjust > Threshold”). After thresholding, fluorescent cells were represented as black areas on a white background. For each image, the area of fluorescent cells was calculated using the command “Analyze > Histogram,” and the number of black pixels was saved as black pixel area (BPA). The black pixel area of the i -th binarized image is abbreviated as BPA[b-img(i)]. All images of a series were processed accordingly, and the pixel numbers obtained were saved as a column in an MS Excel spreadsheet. These values were plotted over time to represent the total cell area per image as a function of time.

To determine the area successively covered by whole cell tracks, binarized images were added using the stack z -projection command (with the setting “minimum intensity”). For the resulting overlay pictures, the number of black pixels (BPA) was counted after projection as described above. Here, we abbreviate the projection of the images 1 to i as Prj(1... i). This procedure was carried out successively yielding Prj(1... i) for $1 \leq i \leq n$, n being the number of images in the stack analyzed. The BPAs of the

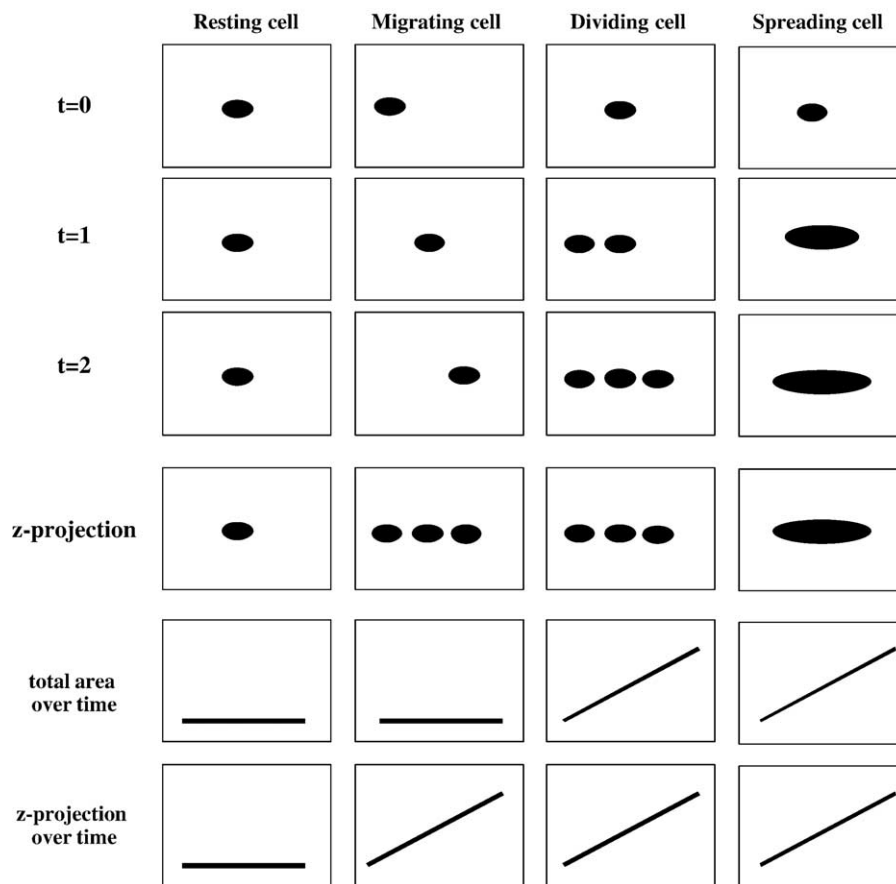


Fig. 1. Principle of the assay. As indicated at the top, schematic time-lapse sequences were shown for cells which rest, migrate, divide or spread. As indicated at the left, the positions of the cells at three consecutive time points are shown. Furthermore, the z-projection of each stack is given as indicated at the left. For each sequence, the diagrams at the bottom represent the total area covered by the cells over time and the area covered by the cells in z-projected images.

consecutive projections were saved as columns in an MS excel spreadsheet. A plot of $BPA[Prj(1 \dots i)]$ against $t = i * \Delta t$ represented the cumulative area covered by cell tracks as a function of the time. This function is called *cumulative cell track area* (measured as the number of black pixels). These time-dependent displacement plots were normalized with respect to the initial BPA in order to obtain the *relative displacement* during time t according to the relationship:

$$RD[t = (i - 1) * \Delta t] = \{BPA[Prj(1 \dots i)] - BPA[b - img(1)]\} / BPA[b - img(1)].$$

As a measure for the steepness of RD over time after normalization for the total cell area and after division by the time interval Δt (h) of the image sequence, the relative displacement index (RDI) was derived according to the formula

$$RDI = RD(\Delta t) / \Delta t = \{BPA[Prj(1 \dots 25)] - BPA[b - img(1)]\} / BPA[b - img(1)] / \Delta t.$$

For curve fitting, data were transferred to an Origin 6.1 (OriginLab) worksheet, plotted and plots were analyzed

using linear regression ($y = a * t$) or interactive curve fitting with a function encompassing linear and square root terms [$y = a * t + \text{sqrt}(b * t)$].

Results

The method proposed here is based on the idea to analyze z-projections of time-lapse sequences in order to measure cell migration. As shown in Fig. 1 in z-projections, a motile cell would successively cover an area growing with time, while the total area in each image will not change over time. A resting cell would cover always the same area. Thus, both the total cell area and the area covered in z-projections will not change over time. On the other hand, dividing cells and spreading cells are also expected to display an increase of the area covered in z-projections, but, in these cases, the total cell area over time would also increase to the same extent. This is in contrast to the migrating cell, where the total cell area is expected to be constant over time. Thus, an increase of the z-projection area which is in excess over the increase of the total cell area indicates cell migration. In the absence of an increase of the total cell area, the quantitative

increase of the area successively covered in *z*-projections is a measure of cell migration.

The feasibility of the method was tested in experiments with B35 neuroblastoma cells. When these cells were seeded at low density and transfected with a GFP expression plasmid, up to 30% of the cells displayed fluorescence after 24 h. Individual fluorescent cells were well separated under these conditions (Fig. 2; panels A and B). Time-lapse video microscopy revealed that these cells were highly motile (data not shown). Transformation of epifluorescence images taken from these cells into a binary black and white representation using a thresholding procedure eliminated any non-cellular background fluorescence and preserved the morphology of cell bodies (Fig. 2; panel C). Some weakly stained finer cell processes, however, were not visible any longer after thresholding. For each image, the total cell area

(BPA[b-img(i)]) was plotted over time (Fig. 2; panel E). The total cell area per image, however, was approximately constant over time or only slightly increasing (Fig. 2; panel E). This indicates that the processes of proliferation or cell spreading were not relevant during the interval of observation. In contrast, a *z*-projected overlay image of all 25 images of a time-lapse sequence after thresholding (Fig. 2; panel D) yielded a global binarized representation of the migration tracks of the population of fluorescent cells monitored in the time-lapse sequence. This area was measured by automatic counting of black pixels in the projected image [Prj(1...*i*)]. The time-dependent cumulative cell track area was determined by iterative application of this procedure (Fig. 2; panel F). In the absence of an increase of the total cell area with time, the increase of the total area covered by *z*-projections of the cells indicates that

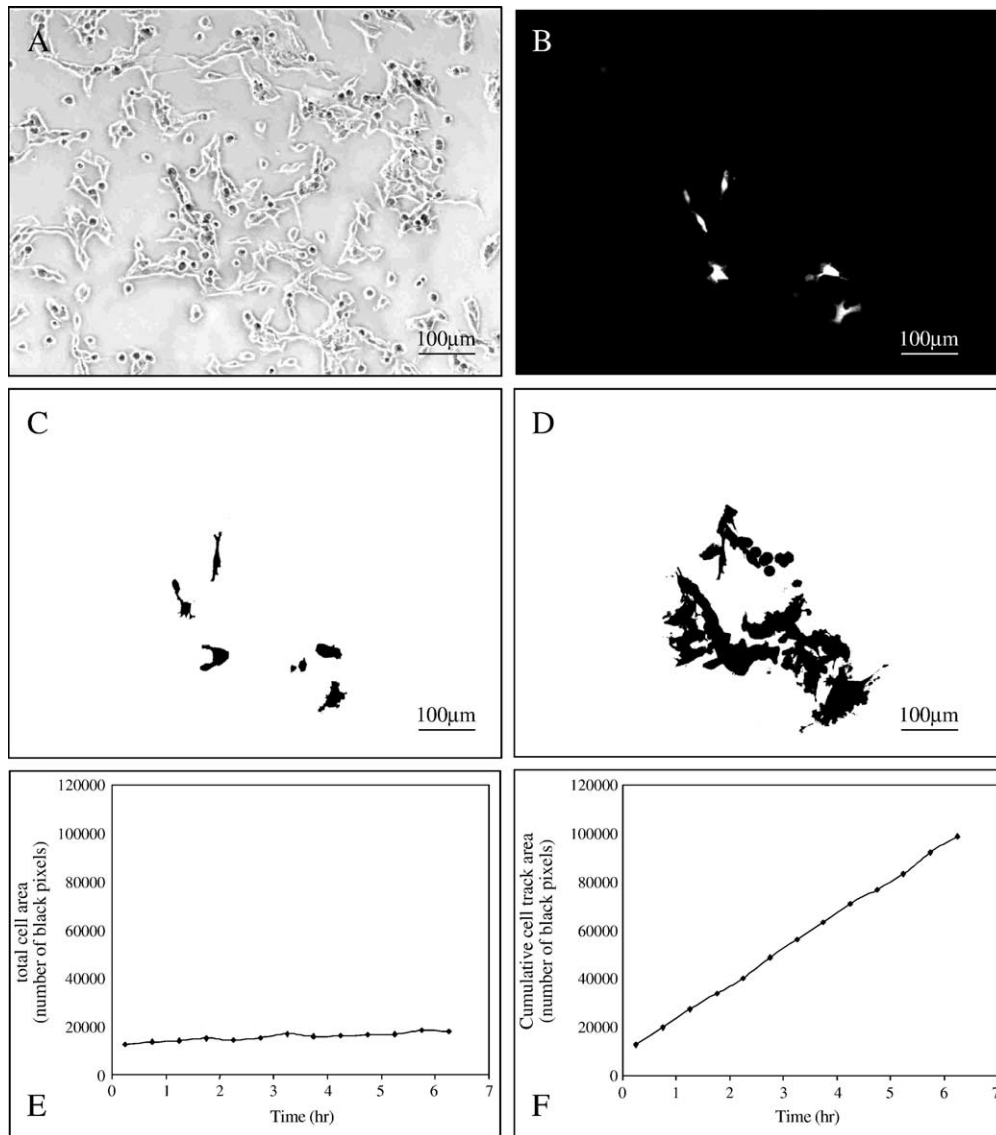


Fig. 2. Migration of B35 neuroblastoma cells transiently expressing GFP. (A) Phase contrast image of the culture. (B) Fluorescence image of the same area. (C) Image after thresholding. (D) Overlay image of the entire time-lapse sequence of 25 images recorded at intervals of 15 min. (E) Plot of total cell area per image over time. Scale bars, 100 μ m. (F) Plot of cumulative cell track area over time, its slope qualifying the mean displacement rate.

the cells migrated during the observation interval. Thus, the areas in the z -projected images represent cell track areas. The plots of the cumulative cell track area increased approximately linearly with time (Fig. 2; panel F). Therefore, the steepness of this function, called relative displacement index (RDI; see Materials and methods section), was used to measure the speed of cell migration. The relative displacement index of B35 neuroblastoma cell under the conditions of our assay was approximately 1 [1/h]. This means that on average each cell has cumulatively covered an area equal to its initial size per hour.

In subsequent experiments, we inhibited cell migration by treatment with cytochalasin B and with methyl-beta-cyclodextrin. Cytochalasin B binds to the plus ends of microfilaments and inhibits actin polymerization which is essential for cell migration. On the other hand, disruption of lipid rafts after extraction of cellular cholesterol with methyl-beta-cyclodextrin [14,15] interferes with cell polarization and inhibits cell migration [16]. Under both conditions, cell growth and/or spreading were negligible, and there was no indication of cytotoxicity as shown by plots of the total cell area (Supplementary materials 1 and 3). Both treatments reduced the slope of the plot of the RD over time considerably (see Fig. 3; panel A and Supplementary materials 1 and 3). Analysis of 8 independent time-lapse sequences for each condition showed a strong inhibition of cell migration in the presence of 10 $\mu\text{g/ml}$ cytochalasin B and an intermediate inhibition after cholesterol extraction with 2 mM methyl-beta-cyclodextrin (Fig. 3; panels A and B). Thus, the effects of these inhibitors of cell migration were clearly detected by our evaluation strategy (Fig. 3). Moreover, when B35 neuroblastoma cells were treated with different concentrations of cytochalasin B, the RDI dropped down to a basal value in a clear dose-response relationship, and half-maximal inhibition of migration was observed at a cytochalasin B concentration of approximately 0.4 $\mu\text{g/ml}$ (Supplementary material 2).

The relative displacement plots over time were mathematically fitted using two models.

For untreated cells, linear regression analysis yielded a Chi-squared value $\chi^2 = 0.04377$ with a squared correlation coefficient $R^2 = 0.9895$ (averaged values from $n = 8$ plots; data not shown). While, for cells treated with cytochalasin B or methyl-beta-cyclodextrin, χ^2 values were in a similar range ($\chi^2 = 0.02526$ for the cytochalasin-B-treated cells and $\chi^2 = 0.02562$ for the methyl-beta-cyclodextrin treated cells; $n = 8$ for each condition; data not shown), the squared correlation coefficients in the latter cases were lower than for untreated cells ($R^2 = 0.87496$ for the cytochalasin B data and $R^2 = 0.93002$ for the methyl-beta-cyclodextrin plots). This was consistent with the observation that the plots observed under these two treatments tended to display slight curvatures (Fig. 3; panel A; Fig. 4; panel A). Based on the assumption that the curvature of these plots was caused by recurrent traces due to multiple random changes of the direction of migration, we included a root term in the

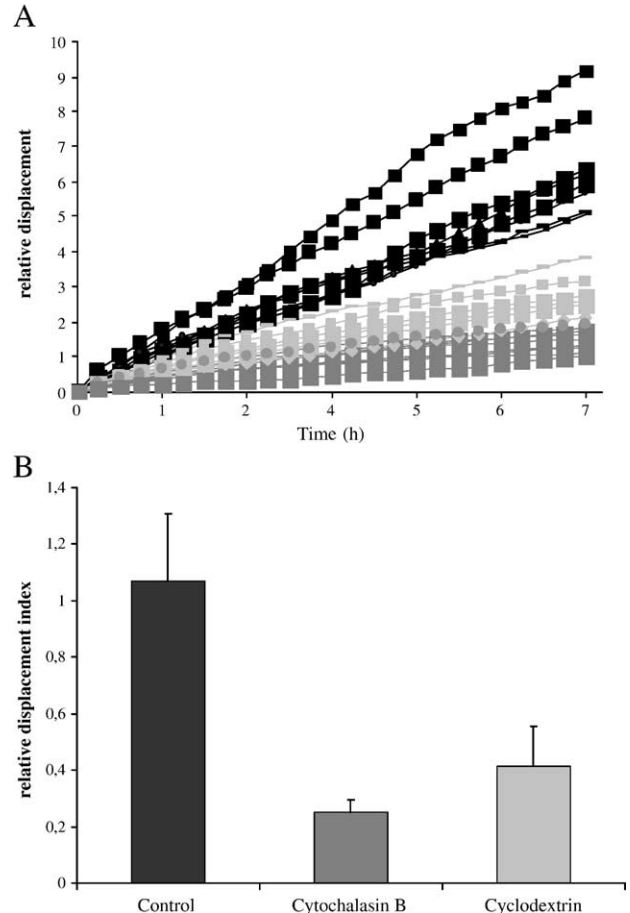


Fig. 3. Comparison of population motility of B35 neuroblastoma cells after normalization. (A) Survey of time-dependent relative displacement plots for untreated cells (black symbols), methyl-beta-cyclodextrin (2 mM) (light gray symbols) and cytochalasin B (10 $\mu\text{g/ml}$) (dark gray symbols). (B) Comparison of relative displacement indices. Bars represent mean values, error bars represent standard deviations ($n = 8$).

function used for curve fitting [$y = a * t + \sqrt{b * t}$]. For the extreme case of no directed motion, i.e. vanishing linear displacement rate ($a = 0$), this curve would correspond to the well-known formula for the mean square displacement of Brownian motion, in our terms written for the non-dimensional quantity, $\text{RD}^2(t) = b * t$, with b denoting a random displacement rate. Thus, the full curve [$y = a * t + \sqrt{b * t}$] can be interpreted as a weighted combination between a linear and a random displacement component. Fitting with this function (Fig. 4; panel A) reduced the χ^2 values for the plots of cytochalasin-B- and methyl-beta-cyclodextrin-treated cells considerably. The global χ^2 values after fitting 8 plots per condition were $\chi^2 = 0.00222$ for cytochalasin-B-treated cells and $\chi^2 = 0.00144$ for methyl-beta-cyclodextrin-treated cells. For untreated cells, the χ^2 was slightly reduced ($\chi^2 = 0.01561$). In this latter case, the linear displacement rate was approximately 5-fold higher than the random displacement rate (Fig. 4; panels B and C). On the other hand, for cells treated with cytochalasin B or methyl-beta-cyclodextrin, respectively, the random displace-

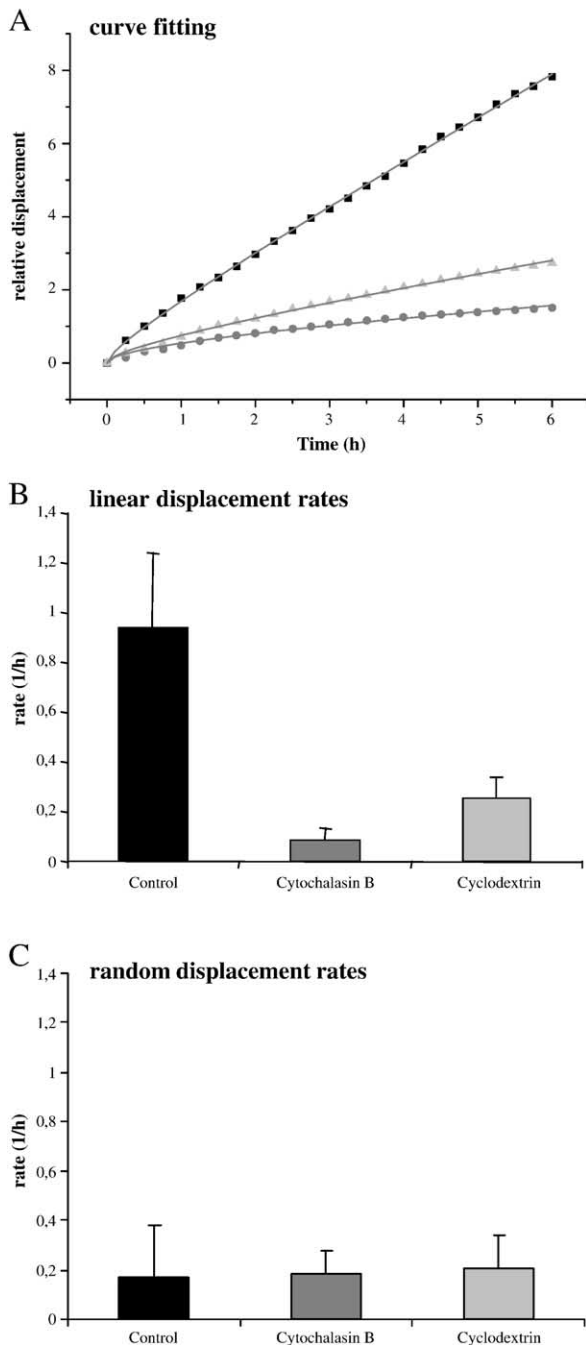


Fig. 4. Fitting of time-dependent relative displacement plots. (A) Fitting of plots with the equation $y = a * t + \text{sqrt}(b * t)$. Plots are shown for untreated cells (black symbols), methyl-beta-cyclodextrin-treated cells (light gray symbols) and cytochalasin-B-treated cells (dark gray symbols). (B and C) Comparison of coefficients of the linear displacement rates (B) and the random displacement rates (C), respectively. Bars represent the mean values for untreated cells (black), methyl-beta-cyclodextrin-treated cells (light gray) and cytochalasin-B-treated cells (dark gray), error bars represent standard deviations ($n = 8$).

ment rates were higher than the linear displacement rates (Fig. 4; panels B and C).

In order to examine the usefulness of our assay to study the migration of other cell types and the compatibility with

other labeling procedures, we used C6 glioma cells after vital staining with the nuclear dye Hoechst 33258. Individual fluorescent nuclei were well separated under these conditions and after thresholding the morphology of nuclei was well preserved (Fig. 5; panel A). The total stained area per image representing the total area covered by nuclei was approximately constant over time (Fig. 5; panel C) excluding cell proliferation. Thus, a z-projected overlay image (Fig. 5; panel B) yielded a global representation of the migration tracks of the nuclei. In the absence of cell proliferation, the increase of the total area covered by z-projections of the cells indicated cell migration during the observation interval (Fig. 5; panel D). The relative displacement index of C6 glioma cells under the conditions of our assay was approximately 1 [1/h]. Treatment of the cells with CB to inhibit cell migration greatly diminished the slope of the relative displacement plot (Supplementary material 4). Thus, our method can be applied to different cell types, and different procedures for fluorescent staining of living cells may be used.

Discussion

In this paper, we describe a general approach to quantify the average motility of cell populations from time-lapse image sequences. Our method is based on the idea to represent the total displacement of cells as the cumulative track area successively covered by the cells due to movements (Fig. 1). At the level of single cells, analysis of area changes due to protrusion or retraction is an established way to monitor cell movements quantitatively [17] when precise area information on individual cells can be obtained, e.g. by microinterferometric techniques [18]. We transferred this approach to measure the motility of B35 neuroblastoma cell populations transiently expressing GFP. This neural cell line was studied previously by others using the transwell assay as a useful model to study the migration of neural precursor cells [19].

In our experiments, the area covered by cells was determined after thresholding of images of fluorescently labeled cells using transient expression of GFP. At the rather stringent cutoff gray values used in this study, this procedure tended to underestimate the true area covered by a cell because the fluorescence within thin protrusions like lamellipodia and filopodia was weak in the low-power magnification images used and was usually below the cutoff threshold which was necessary to exclude any nonspecific background (Fig. 2; panels A–C). Thus, area changes measured are likely to represent movements of the cell bodies. This implies that our assay is biased towards the detection of cell migration, whereas non-directed fluctuations of the cell margins are incompletely detected. The increase of the track area of the B35 cell population (Fig. 2; panel D) was approximately linear with time (Fig. 2; panel E). Accordingly, regression analysis yielded good fits of the

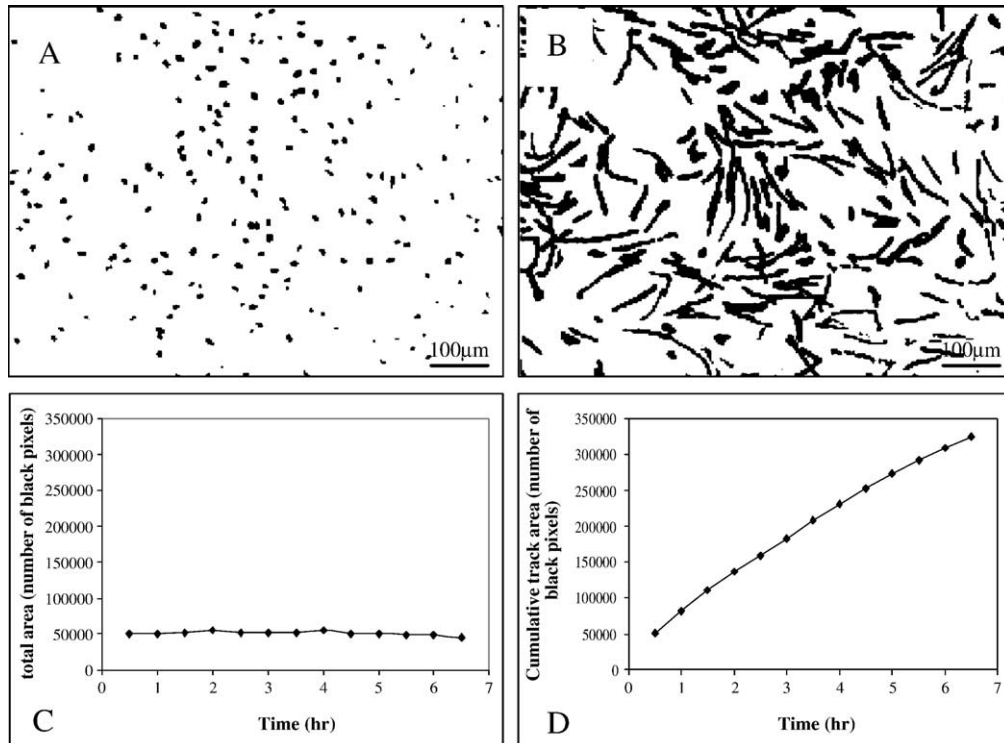


Fig. 5. Migration of C6 glioma cells after vital staining with Hoechst 33258. (A) Image after thresholding. (B) Overlay image of the entire time-lapse sequence of 25 images recorded at intervals of 15 min. (C) Plot of total stained area per image over time. (D) Plot of cumulative track area over time. Scale bars, 100 μm.

experimental data. The linear increase of the displacement plots indicates that recurrence of migration tracks due to multiple random changes of the direction of migration is of minor significance under control conditions. Thus, we conclude that the directional persistence time of migration of B35 neuroblastoma cells is of the same magnitude or larger than the time-scale of our assay for most cells (6 h). Therefore, in the absence of relevant cell growth (Fig. 2; panel F), a simple relative displacement index defined as $RDI = [(area\ of\ projection - area\ of\ first\ image) / area] / \Delta t$ quantitatively describes the cell migration of these populations. As expected, inhibition of actin polymerization with cytochalasin B markedly reduced cell migration (Figs. 3 and 4; Supplementary materials 1 and 2), and our assay was useful to determine dose–response relationships for agents interfering with cell migration (Supplementary material 2). Furthermore, we found that extraction of cholesterol with methyl-beta-cyclodextrin markedly reduced cell migration (Figs. 3 and 4; Supplementary material 3). This treatment interferes with cell polarization since the sorting of cholesterol-rich microdomains of the plasma membrane is necessary to establish and maintain membrane organization into front and rear edges, e.g. of adenocarcinoma cells [20] and T-cells [16]. Our observations show that cholesterol-rich microdomains are essential for directed migration of B35 neuroblastoma cells.

Interestingly, displacement plots frequently displayed convex curvatures when cells were observed in the presence of cytochalasin B or after cholesterol extraction (Figs. 3 and

4; panel A). Therefore, satisfactory curve fitting was achieved only after including a root term (Fig. 4; panel A). Furthermore, fitting of the displacement plots of untreated cells with this convex model function yielded a χ^2 value which was slightly smaller than that obtained with linear regression. As detailed above (see Results section), the presence of a root term is indicative of self recurrent tracks. An extreme case of such a self recurrent track would occur when a cell turns and moves the same way back, but even this cell would contribute to the increase of the RD until it turns. In general, a low directional persistence of migration with respect to the time-scale of the assay could cause self recurrent tracks. On the other hand, tracks from different cells may cross each other more frequently depending on cell density and/or on the mutual orientation of tracks. Therefore, it is important to use a low to moderate density of labeled cells during the assay because otherwise saturation of black pixels could occur. This is easily adjusted when transient transfection with a GFP-expressing plasmid is used (Fig. 2). However, even after vital staining of C6 glioma cells, there was no saturation when the length of the time-lapse sequence was kept within reasonable limits (Fig. 5). Moreover, fluorescence fading may also gradually reduce the slope of RD plots over time. Under the conditions employed, however, photobleaching was not relevant, as indicated by a comparison of the distributions of pixel intensities between the different images of a sequence. If photobleaching could be not avoided by adjusting the time interval between the images, it may be possible to

adjust the threshold level used for image binarizing in order to compensate for this, e.g. by using a multiplication factor to keep the summed pixel intensity level for any single cell (before binarizing) constant throughout the image sequence. When attention is paid to these limitations, the relative displacement index as defined above is a reliable conservative estimate for the motile behavior of cells.

Our method was easily programmed as a macro in the freely available Image J software package. Mathematical fitting of the displacement plots was carried out in Origin 6.1 using a heuristic model function (see above). However, other program packages like Sigma Plot may be used, and the model function may be changed. The simplicity of the operations used allowed automated rapid evaluation of one hundred and more image sequences typically obtained in a single experiment. The fast speed of evaluation was possible because for our algorithm it is not necessary to recognize individual cells as separate entities. Thus, our test may be applied for the analysis of images which contain high cell numbers if enough area is left to be covered by movements. Furthermore, it is suited to examine quickly large numbers of conditions as it is necessary in screening experiments or when dose–response relationships are examined.

Since the relative displacement plots represent the motility of up to 40 cells per image sequence, the normalized motility index derived from such plots is a rather robust statistical measure reflecting the average migration in cell populations. As in the Boyden transwell assay, our approach does not yield a dispersion measure which represents the distribution of different migration types and velocities within the cell population. In order to obtain information on individual cells, it would be possible to divide the images into appropriately sized rectangular segments which contain single fluorescent cells. This would allow to calculate migration indices for single cells and to determine the distribution of this index in the cell population. Though possible, such an analysis was not the primary intention of our method since subtle cell tracking algorithms for individual cells are available (e.g. MTrack2). These programs, however, need more computation time, and the output is a large set of numbers describing the individual cell tracks. From these tracks, mean values and variances are then derived. In comparison to these algorithms, it is an advantage of our method that a single number qualifying the motility of a whole cell population is assigned very rapidly. Using this number as a criterion, it is possible to screen and quickly categorize a large number of different conditions. In contrast to the Boyden chamber assay, however, the same raw data, i.e. image sequences, can be used directly for a secondary evaluation using more elaborate and time-consuming methods, if a global alteration of cell migration is detected by our method.

It is important to notice that our method is not limited to the examination of B35 neuroblastoma cells and that other labeling procedures than expression of GFP may be

used. When we applied the method to measure the migration of C6 glioma cells after vital staining with Hoechst dye, our evaluation lead to comparable results as in the case of B35 neuroblastoma cells (Fig. 5). It is obvious, however, that after nuclear staining the assay measures major movements of the center of mass of the cells while movements of cell protrusions which do not lead to shift of the nucleus are not detected. Interestingly, under these conditions, the tracks were rather line-shaped with smooth contours (Fig. 5; panel B). This may open the possibility to study the shapes of individual tracks and to obtain information on the heterogeneity of migration behavior in cell populations. Meanwhile, we have applied the method successfully in experiments with other cell types like keratinocyte cell lines (Dai, Kappler and Magin; unpublished observations) or even cultures of brain slices (Borowski, Alt, Kappler and Schilling, in preparation). In all cases, relative displacement indices could be measured as detailed above for B35 neuroblastoma cells and C6 glioma cells after appropriate adjustment of the sampling rate. Furthermore, we have successfully used the method in overexpression and knockdown experiments to study gene function [21]. We conclude that the calculation of the time-dependent displacement of cell populations is a versatile tool to obtain quantitative information on cell migration in a wide variety of cell culture models.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.yexcr.2005.09.013](https://doi.org/10.1016/j.yexcr.2005.09.013).

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